CSC5041: Assignment 2
Deadline: December 10, 2026 (23:59)
Total marks: 15

Question 1 (5 marks)

Consider the relation schema $R(P,Q,O,S,T,U,V,W,X)$ and the set of functional dependencies

$$F = \{PS \to Q,\ QS \to U,\ QT \to W,\ PT \to SW,\ PW \to T,\ PTW \to O\}.$$

Assume that all attribute values are atomic, i.e. R is in 1NF.

(1) Check whether $PQ \to U$ is logically implied by F (show the attribute closure $\{P,Q\}^+$), and find *all* candidate keys of R. *(1 mark)*

(2) Determine the highest normal form of R with respect to F. Justify your answer. *(1 mark)*

(3) Find a minimal cover F_m for F, and give a *lossless, dependency-preserving* decomposition of R into 3NF. *(1 mark)*

(4) With respect to F, does the decomposition $R_1 = \{P,Q,S,X,O\}$, $R_2 = \{Q,S,U,W\}$, $R_3 = \{Q,T,O,V\}$ of R have the lossless-join property, and does it preserve the dependencies? Justify both answers. *(1 mark)*

(5) Give a decomposition of R into BCNF, and state whether your decomposition preserves the dependencies in F. *(1 mark)*

Question 2 (5 marks)

Consider the schedule shown in Table 1. Here $R(*)$ and $W(*)$ stand for 'read' and 'write', respectively; T_1, T_2, T_3 are three transactions; and $t_1, t_2, \ldots, t_{12}$ represent successive time slots.

Table 1: Schedule of the three transactions.

	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}	t_{12}
T_1	$R(x)$			$R(y)$				$W(x)$			$W(y)$	
T_2			$R(y)$		$R(z)$				$W(y)$			$W(z)$
T_3		$R(z)$				$R(x)$	$W(z)$			$W(x)$		

Note. Each transaction starts at the time slot of its first operation and commits immediately after its last operation. For parts (2) and (3), the time slots fix only the *order* in which operations and lock requests are issued; a transaction that is blocked waits until it can proceed, so its later operations may be delayed beyond their nominal time slots. A blocked transaction cannot commit: it reaches its commit point only after its last operation has actually executed, and until then it keeps every lock it already holds.

Give and justify your answers to the following.

(1) Is the schedule *conflict-serializable*? If it is not, draw the precedence graph and identify a cycle; otherwise, give an equivalent serial schedule. *(1 mark)*

(2) Use **rigorous two-phase locking** to add the necessary lock and unlock actions to each transaction:

- before each $R_i(A)$, T_i requests a shared lock $S_i(A)$;
- before each $W_i(A)$, T_i requests (or upgrades to) an exclusive lock $X_i(A)$;
- T_i releases *all* its locks, denoted U_i, only when it commits;
- lock requests are issued in the order in which the operations appear in Table 1.

Write down the resulting sequence for each transaction and state where its growing phase ends. *(2 marks)*

(3) Suppose the transactions now try to carry out the operations in exactly the order of Table 1 under the locking scheme of part (2). What happens? Draw the *wait-for graph* at that point and justify your conclusion. *(2 marks)*

Question 3 (5 marks)

Consider the following *page reference string*, where P_i denotes the page with identifier i:

$$P_1,\ P_2,\ P_3,\ P_4,\ P_3,\ P_4,\ P_1,\ P_2,\ P_4,\ P_5,\ P_2,\ P_1,\ P_3,\ P_2,\ P_1,\ P_4.$$

The buffer pool has 3 frames and is initially empty. A request for a page already resident in the pool is a *hit* and causes no disk I/O; otherwise a *page fault* occurs and the requested page is loaded. Whenever the pool is full and a page fault occurs, exactly one resident page is evicted according to the policy in question. For **MRU**, the victim is the page whose *most recent access* is the latest one, and every access — including a hit — updates this recency order.

(1) Sketch the replacement process under the Least Recently Used (LRU) policy: give the buffer contents after each request, the page evicted, and hit/miss. *(1 mark)*

(2) Do the same for the Most Recently Used (MRU) policy, as defined above. *(1 mark)*

(3) Do the same for the First In First Out (FIFO) policy; for FIFO, a hit does not change the arrival order. *(1 mark)*

(4) State the total number of page faults under each of the three policies. Compare LRU and FIFO: do the two policies perform differently on this reference string? Explain your answer by referring to the individual requests where their behaviour differs. *(2 marks)*

Assignment Submission

- You must submit an electronic copy of your answers to the above questions to the course website.
- We only accept .pdf files.
- Please name your files in the following format: `226010XXX-Assg2.pdf`, replacing `226010XXX` with your own student ID.

Note:

1) If you have problems relating to your submission, please contact the teaching assistant on WeChat or Email.
2) All submissions will be checked for plagiarism. The university regards plagiarism as a form of academic misconduct and has very strict rules. Not knowing the rules will not be considered a valid excuse when you are caught.

Late Submission Penalty

- A hefty 5% of your final mark will be deducted for each additional day (24hr) after the specified submission time and date.
- Submissions that are more than three days late will not be marked.

Use of AI Tool

- You may use AI tools to help you understand a question or to get a hint.
- Submitting work that was produced wholly by an AI will be treated as academic misconduct.